Week 6: Product Review and Rating Management System

Scenario:

The e-commerce company wants to collect customer feedback about purchased products. Customers can provide ratings and reviews after completing their orders. The system stores customer opinions, analyzes product performance, and identifies highly rated products. The Product Review and Rating Module is connected with the existing Customer and Product modules.

Tasks to be Done

1. Create Review and Rating Tables

A review table stores the review ID, customer ID, product ID, rating, review text, and review date. A primary key uniquely identifies each review, and foreign keys connect reviews to valid customers and products.

```
SQL> CREATE TABLE Review(ReviewID VARCHAR2(7),
  2  CustomerID VARCHAR2(7),
  3  ProductID VARCHAR2(7),
  4  Rating NUMBER(1),
  5  ReviewTest VARCHAR2(50),
  6  ReviewDate DATE,
  7  PRIMARY KEY(ReviewID),
  8  FOREIGN KEY(CustomerID) REFERENCES Customer(Customer_ID),
  9  FOREIGN KEY(ProductID) REFERENCES PRoduct(Product_ID));

Table created.

SQL> desc Review
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------
 REVIEWID                                  NOT NULL VARCHAR2(7)
 CUSTOMERID                                         VARCHAR2(7)
 PRODUCTID                                          VARCHAR2(7)
 RATING                                             NUMBER(1)
 REVIEWTEST                                         VARCHAR2(50)
 REVIEWDATE                                         DATE
```

Store Customer Feedback and Ratings

Reviews record a customer's opinion about a purchased product. Ratings represent satisfaction on a five-star scale.

1. Add reviews for purchased products.

```
SQL> INSERT INTO REVIEW
  2  (REVIEWID, CUSTOMERID, PRODUCTID, RATING, REVIEWTEST, REVIEWDATE)
  3  VALUES
  4  ('REV0001', 'CUST000001', 'P000001', 5, 'Excellent laptop performance', DATE '2026-01-2

1 row created.

SQL>
SQL> INSERT INTO REVIEW
  2  (REVIEWID, CUSTOMERID, PRODUCTID, RATING, REVIEWTEST, REVIEWDATE)
  3  VALUES
  4  ('REV0002', 'CUST000001', 'P000002', 4, 'Good wireless mouse', DATE '2026-01-21');

1 row created.

SQL>
SQL> INSERT INTO REVIEW
  2  (REVIEWID, CUSTOMERID, PRODUCTID, RATING, REVIEWTEST, REVIEWDATE)
  3  VALUES
  4  ('REV0003', 'CUST000002', 'P000004', 5, 'Very good sound quality', DATE '2026-02-02');

1 row created.

SQL>
SQL> INSERT INTO REVIEW
  2  (REVIEWID, CUSTOMERID, PRODUCTID, RATING, REVIEWTEST, REVIEWDATE)
  3  VALUES
  4  ('REV0004', 'CUST000002', 'P000006', 4, 'Good quality USB cable', DATE '2026-02-03');

1 row created.

SQL>
SQL> INSERT INTO REVIEW
  2  (REVIEWID, CUSTOMERID, PRODUCTID, RATING, REVIEWTEST, REVIEWDATE)
  3  VALUES
  4  ('REV0005', 'CUST000003', 'P000005', 5, 'Excellent smartphone', DATE '2026-02-15');

1 row created.
```

2. Store customer ratings from 1 to 5 stars

A rating is a numeric score representing customer satisfaction. The CHECK constraint ensures only values from 1 to 5 are accepted.

```
SQL> SELECT * FROM Review;

REVIEWI CUSTOMERID PRODUCT     RATING
------- ---------- -------- ----------
REVIEWTEST                                         REVIEWDAT
-------------------------------------------------- ---------
REV0001 CUST000001 P000001          5
Excellent laptop performance                       20-JAN-26

REV0002 CUST000001 P000002          4
Good wireless mouse                                21-JAN-26

REV0003 CUST000002 P000004          5
Very good sound quality                            02-FEB-26


REVIEWI CUSTOMERID PRODUCT     RATING
------- ---------- -------- ----------
REVIEWTEST                                         REVIEWDAT
-------------------------------------------------- ---------
REV0004 CUST000002 P000006          4
Good quality USB cable                             03-FEB-26

REV0005 CUST000003 P000005          5
Excellent smartphone                               15-FEB-26

REV0006 CUST000004 P000008          4
Useful smart watch                                 28-FEB-26


REVIEWI CUSTOMERID PRODUCT     RATING
------- ---------- -------- ----------
REVIEWTEST                                         REVIEWDAT
-------------------------------------------------- ---------
REV0007 CUST000004 P000006          5
Cable works perfectly                              01-MAR-26

REV0008 CUST000005 P000010          4
Good tablet for daily use                          12-MAR-26

REV0009 CUST000006 P000009          5
Clear and comfortable headphones                   25-MAR-26
```

Ensure this by adding a constraint

```
ALTER TABLE Review ADD CONSTRAINT CHK_Rate CHECK (Rating >= 1 AND Rating <= 5);
```

```
SQL> ALTER TABLE Review ADD CONSTRAINT CHK_Rate CHECK(Rating >= 1 AND Rating <= 5);

Table altered.
```

3. Update review comments if required

UPDATE changes the text of an existing review. The WHERE condition targets the required review ID.



4. Remove inappropriate or invalid reviews

DELETE removes a selected review record when it is invalid or violates the review policy.



4. Retrieve Product Review Details

SQL queries can retrieve feedback for one product, show customer names with their comments, identify products with many reviews, display recent feedback, and filter reviews by rating.

1. Display all reviews for a product

This query filters review records by the selected ProductID.

```
SQL> SELECT * FROM Review WHERE ProductID ='P000012';

REVIEWI CUSTOMERID PRODUCT     RATING
------- ---------- ------- ----------
REVIEWTEST                                          REVIEWDAT
--------------------------------------------------- ---------
REV0012 CUST000008 P000012          4
Reliable storage device                             30-APR-26

REV0014 CUST000010 P000012          5
Very useful external drive                          28-MAY-26
```

2. Display customer name with their reviews

JOIN combines customer information with the reviews submitted by each customer.

```
SQL> SELECT CONCAT(c.First_Name,c.Last_Name) AS Customer_Name,
  2  R.ReviewID,R.ProductID,R.Rating,R.ReviewTest
  3  FROM Customer c JOIN Review R
  4  ON c.Customer_ID = R.CustomerID;

CUSTOMER_NAME                          REVIEWI PRODUCT     RATING
-------------------------------------- ------- ------- ----------
REVIEWTEST
--------------------------------------------------
ArjunKumar                             REV0001 P000001          5
Excellent laptop performance

ArjunKumar                             REV0002 P000002          4
Good wireless mouse

PriyaSharma                            REV0003 P000004          5
Very good sound quality


CUSTOMER_NAME                          REVIEWI PRODUCT     RATING
-------------------------------------- ------- ------- ----------
REVIEWTEST
--------------------------------------------------
PriyaSharma                            REV0004 P000006          4
Good quality USB cable

RahulMenon                             REV0005 P000005          5
Excellent smartphone

AnanyaIyer                             REV0006 P000008          4
Useful smart watch
```

3. Find products having maximum reviews

COUNT calculates the number of reviews for each product. ORDER BY in descending order places products with the most reviews first.

```
SQL> SELECT
  2      P.PRODUCT_NAME,
  3      COUNT(R.REVIEWID) AS REVIEW_COUNT
  4  FROM PRODUCT P
  5  JOIN REVIEW R
  6      ON P.PRODUCT_ID = R.PRODUCTID
  7  GROUP BY P.PRODUCT_ID, P.PRODUCT_NAME
  8  ORDER BY REVIEW_COUNT DESC;

PRODUCT_NAME                                       REVIEW_COUNT
-------------------------------------------------- ------------
USB Cable                                                     3
Smart Watch                                                   2
External Hard Disk                                            2
Wireless Earbuds                                              2
Headphones                                                    2
Laptop Stand                                                  1
Tablet                                                        1
Smartphone                                                    1
Laptop                                                        1
Bluetooth Speaker                                             1
Wireless Mouse                                                1

11 rows selected.
```

4. Display recent customer feedback

Sorting by review date in descending order shows the newest feedback first. The query can return the five most recent reviews.

```
SQL> SELECT
  2      P.PRODUCT_ID,
  3      P.PRODUCT_NAME,
  4      ROUND(AVG(R.RATING), 2) AS AVERAGE_RATING
  5  FROM PRODUCT P
  6  JOIN REVIEW R
  7      ON P.PRODUCT_ID = R.PRODUCTID
  8  GROUP BY P.PRODUCT_ID, P.PRODUCT_NAME
  9  HAVING AVG(R.RATING) = (
 10      SELECT MAX(AVG_RATING)
 11      FROM (
 12          SELECT AVG(RATING) AS AVG_RATING
 13          FROM REVIEW
 14          GROUP BY PRODUCTID
 15      )
 16  );

PRODUCT PRODUCT_NAME                                  AVERAGE_RATING
------- -------------------------------------------- --------------
P000001 Laptop                                                    5
P000004 Bluetooth Speaker                                         5
P000005 Smartphone                                                5
```

5. Retrieve reviews with ratings above 4

The condition Rating > 4 selects five-star reviews because the valid rating range is 1 to 5.

```
SQL> SELECT
  2      P.PRODUCT_ID,
  3      P.PRODUCT_NAME,
  4      COUNT(R.REVIEWID) AS REVIEW_COUNT
  5  FROM PRODUCT P
  6  JOIN REVIEW R
  7      ON P.PRODUCT_ID = R.PRODUCTID
  8  GROUP BY P.PRODUCT_ID, P.PRODUCT_NAME
  9  ORDER BY REVIEW_COUNT DESC;

PRODUCT PRODUCT_NAME                                       REVIEW_COUNT
------- -------------------------------------------------- ------------
P000006 USB Cable                                                     3
P000008 Smart Watch                                                   2
P000012 External Hard Disk                                            2
P000015 Wireless Earbuds                                              2
P000009 Headphones                                                    2
P000014 Laptop Stand                                                  1
P000010 Tablet                                                        1
P000005 Smartphone                                                    1
P000001 Laptop                                                        1
P000004 Bluetooth Speaker                                             1
P000002 Wireless Mouse                                                1

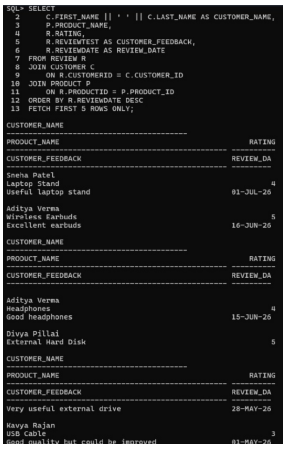
11 rows selected.
```

Example query: calculate average product ratings

Aggregate functions summarize the ratings submitted for products.

1. Calculate average rating for each product

AVG calculates the mean rating for each product. ROUND can display the result to two decimal places.



2. Count the number of reviews per product

COUNT calculates the number of reviews for each product and helps identify which products receive the most feedback.

```
SQL> SELECT
  2       P.PRODUCT_NAME,
  3       COUNT(R.REVIEWID) AS NUMBER_OF_REVIEWS,
  4       ROUND(AVG(R.RATING), 2) AS AVERAGE_RATING
  5  FROM PRODUCT P
  6  LEFT JOIN REVIEW R
  7       ON P.PRODUCT_ID = R.PRODUCTID
  8  GROUP BY P.PRODUCT_ID, P.PRODUCT_NAME
  9  ORDER BY AVERAGE_RATING DESC NULLS LAST;
```

3. Find highest-rated products

The highest-rated product is the product with the maximum average rating. If several products share the same maximum average, all should be reported.

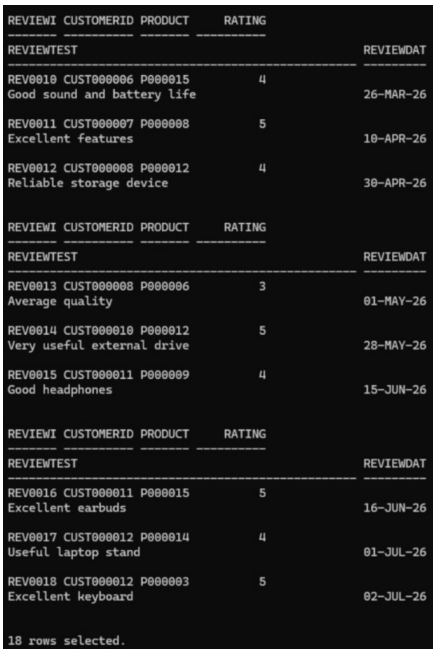
```
SQL> SELECT
  2       P.PRODUCT_ID,
  3       P.PRODUCT_NAME,
  4       ROUND(AVG(R.RATING), 2) AS AVERAGE_RATING
  5  FROM PRODUCT P
  6  JOIN REVIEW R
  7       ON P.PRODUCT_ID = R.PRODUCTID
  8  GROUP BY P.PRODUCT_ID, P.PRODUCT_NAME
  9  HAVING AVG(R.RATING) > 4
 10  ORDER BY AVERAGE_RATING DESC;

PRODUCT PRODUCT_NAME                                       AVERAGE_RATING
------- -------------------------------------------------- --------------
P000005 Smartphone                                                      5
P000004 Bluetooth Speaker                                               5
P000001 Laptop                                                          5
P000012 External Hard Disk                                            4.5
P000009 Headphones                                                    4.5
P000015 Wireless Earbuds                                              4.5
P000008 Smart Watch                                                   4.5

7 rows selected.
```

4. Identify products with average rating above 4

GROUP BY collects reviews by product, and HAVING filters the groups to show products whose average rating exceeds 4.



```
REVIEWI CUSTOMERID PRODUCT     RATING
------- ---------- ------- ----------
REVIEWTEST                                         REVIEWDAT
-------------------------------------------------- ---------
REV0010 CUST000006 P000015          4
Good sound and battery life                        26-MAR-26

REV0011 CUST000007 P000008          5
Excellent features                                 10-APR-26

REV0012 CUST000008 P000012          4
Reliable storage device                            30-APR-26


REVIEWI CUSTOMERID PRODUCT     RATING
------- ---------- ------- ----------
REVIEWTEST                                         REVIEWDAT
-------------------------------------------------- ---------
REV0013 CUST000008 P000006          3
Average quality                                    01-MAY-26

REV0014 CUST000010 P000012          5
Very useful external drive                         28-MAY-26

REV0015 CUST000011 P000009          4
Good headphones                                    15-JUN-26


REVIEWI CUSTOMERID PRODUCT     RATING
------- ---------- ------- ----------
REVIEWTEST                                         REVIEWDAT
-------------------------------------------------- ---------
REV0016 CUST000011 P000015          5
Excellent earbuds                                  16-JUN-26

REV0017 CUST000012 P000014          4
Useful laptop stand                                01-JUL-26

REV0018 CUST000012 P000003          5
Excellent keyboard                                 02-JUL-26

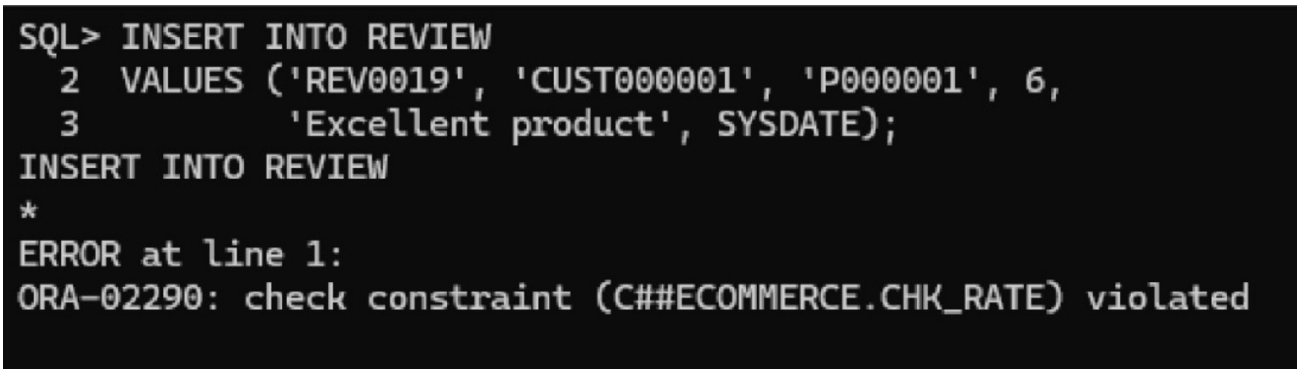

18 rows selected.
```

6. Identify Highly Rated Products

Product-rating reports help compare customer satisfaction, identify popular products, and discover items that may need improvement.

Report 1: Product Rating Analysis

Display: Product name, number of reviews, and average rating.



```
SQL> INSERT INTO REVIEW
  2  VALUES ('REV0019', 'CUST000001', 'P000001', 6,
  3          'Excellent product', SYSDATE);
INSERT INTO REVIEW
*
ERROR at line 1:
ORA-02290: check constraint (C##ECOMMERCE.CHK_RATE) violated
```

Report 2: Customer Feedback Analysis

This report identifies the most reviewed products, highly rated products, and products requiring improvement. Review counts measure feedback volume; average ratings summarize satisfaction.

Most reviewed products

```
SQL> SELECT
  2       P.PRODUCT_NAME,
  3       COUNT(R.REVIEWID) AS NUMBER_OF_REVIEWS,
  4       ROUND(AVG(R.RATING), 2) AS AVERAGE_RATING
  5  FROM PRODUCT P
  6  LEFT JOIN REVIEW R
  7       ON P.PRODUCT_ID = R.PRODUCTID
  8  GROUP BY P.PRODUCT_ID, P.PRODUCT_NAME
  9  ORDER BY AVERAGE_RATING DESC NULLS LAST;
```

Highly rated products

```
SQL> SELECT
  2       P.PRODUCT_ID,
  3       P.PRODUCT_NAME,
  4       ROUND(AVG(R.RATING), 2) AS AVERAGE_RATING
  5  FROM PRODUCT P
  6  JOIN REVIEW R
  7       ON P.PRODUCT_ID = R.PRODUCTID
  8  GROUP BY P.PRODUCT_ID, P.PRODUCT_NAME
  9  HAVING AVG(R.RATING) > 4
 10  ORDER BY AVERAGE_RATING DESC;

PRODUCT PRODUCT_NAME                                    AVERAGE_RATING
------- ---------------------------------------------- --------------
P000005 Smartphone                                                  5
P000004 Bluetooth Speaker                                           5
P000001 Laptop                                                      5
P000012 External Hard Disk                                        4.5
P000009 Headphones                                                4.5
P000015 Wireless Earbuds                                          4.5
P000008 Smart Watch                                               4.5

7 rows selected.
```

Products requiring improvement

Products with low average ratings or repeated negative comments may need further investigation. A query using HAVING AVG(Rating) < 3 can identify low-average products with reviews.

```
SQL> DELETE FROM REVIEW
  2  WHERE REVIEWID = 'REV0018';

1 row deleted.
```

Report 3: Rating Distribution

The rating distribution counts how many reviews have each star value and how many products have low average ratings.

Number of 5-star ratings

```
SQL> SELECT COUNT(*) AS LOW_RATED_PRODUC
  2  FROM (
  3      SELECT PRODUCTID
  4      FROM REVIEW
  5      GROUP BY PRODUCTID
  6      HAVING AVG(RATING) < 3
  7  );

LOW_RATED_PRODUCTS
------------------
                 0
```

Number of 4-star ratings

```
SQL> SELECT
  2      P.PRODUCT_NAME,
  3      ROUND(AVG(R.RATING), 2) AS AVERAGE_RATING
  4  FROM PRODUCT P
  5  JOIN REVIEW R
  6      ON P.PRODUCT_ID = R.PRODUCTID
  7  GROUP BY P.PRODUCT_ID, P.PRODUCT_NAME
  8  HAVING AVG(R.RATING) > 4
  9  ORDER BY AVERAGE_RATING DESC;

PRODUCT_NAME                                       AVERAGE_RATING
-------------------------------------------------- --------------
Smartphone                                                      5
Bluetooth Speaker                                               5
Laptop                                                          5
External Hard Disk                                            4.5
Headphones                                                    4.5
Wireless Earbuds                                              4.5
Smart Watch                                                   4.5

7 rows selected.
```

Number of low-rated products

```
SQL> DELETE FROM REVIEW
  2  WHERE REVIEWID = 'REV0018';

1 row deleted.
```

Complete Rating

A complete rating summary includes the counts of 1-, 2-, 3-, 4-, and 5-star reviews, review counts per product, and the average rating for each product.

```
SQL> SELECT p.Product_Name, Count(R.Rating) AS Count_Rating
  2  FROM Product p JOIN Review R
  3  ON p.Product_ID = R.ProductID
  4  GROUP BY p.Product_Name;

PRODUCT_NAME                                       COUNT_RATING
-------------------------------------------------- ------------
Laptop                                                        1
Wireless Mouse                                                1
Bluetooth Speaker                                             1
Smartphone                                                    1
USB Cable                                                     3
Smart Watch                                                   2
Headphones                                                    2
Tablet                                                        1
External Hard Disk                                            2
Laptop Stand                                                  1
Wireless Earbuds                                              2

11 rows selected
```

Definitions and Conclusion

Review: Written feedback submitted by a customer about a product.

Rating: A numerical score that expresses customer satisfaction, here measured from 1 to 5 stars.

Primary key: A field that uniquely identifies each row in a table.

Foreign key: A field that references a key in another table to maintain valid relationships.

CHECK constraint: A rule that restricts values to those satisfying a specified condition.

JOIN: Combines related rows from different tables using a matching condition.

Aggregate functions: Functions such as COUNT and AVG that summarize multiple rows.

GROUP BY: Groups rows so aggregate calculations can be performed for each group.

HAVING: Filters grouped results after aggregate calculations.

Conclusion: The Product Review and Rating Management System stores customer feedback, validates ratings, and supports review retrieval and product analysis. SQL constraints protect data quality, while joins and aggregate functions help identify the most reviewed, highest-rated, and low-rated products.